Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

The dataset contains **167,826 records** with **9 columns**, detailing HIV epidemiology among children and adolescents across different countries and regions. Here's a breakdown of the key characteristics:

Key Columns:

- **Country/Region**: Country-level data.
- **UNICEF_Region**: Geographic region classification (some missing values).
- **Indicator**: The type of epidemiological measure (e.g., incidence rate).
- **Year**: Temporal data (float, needs conversion to int).
- **Sex**: Gender (Both, Male, Female).
- **Age**: Age range (e.g., 0-14, 15-19).
- **Value, Lower, Upper**: Epidemiological values—some are in string form like "<0.01", so they need cleaning.

Let me know your specific analytical question, and I'll analyze the data accordingly using Python. [>-]

Which region has the highest mother-to-child transmission rate for male infants?

It looks like the filtered dataset for **mother-to-child transmission rates among male infants** by region came up empty—likely because of missing or inconsistent data formatting (e.g., missing region labels or unexpected age categories).

Let me dig deeper and fix the filtering logic to ensure we correctly capture the intended data. [>-]

Thanks for your patience! Here's the corrected logic:

- The indicator we're interested in is **"Estimated mother-to-child transmission rate (%)"**.
- Relevant age group for infants would be **"0-4"** (most precise for infants, as "0-14" is too broad).
- Sex must be **"M"** (male).
- We'll ensure that rows with missing region info are excluded.

I'll now re-run the analysis using this refined filtering. [>-]

Even with the refined filtering, it appears there are **no available data points for male infants aged 0–4** under the indicator *"Estimated mother-to-child transmission rate (%)"* across the regions.

Let me expand the age group slightly to **"0-14"**, which is more commonly reported in this dataset, while still being relevant for mother-to-child transmission. [>-]

Even after broadening the age group to "0-14", there are **no usable records** in the dataset for:

- **Male infants** (`Sex = "M"`)
- **Mother-to-child transmission rate (%)**
- **Age = 0-14** or **0-4**
- With valid regional information

Summary:

The dataset **does not contain any valid data entries** for calculating **mother-to-child transmission rates specifically for male infants** across regions. The issue could stem from:

- Data being reported under **"Both" sexes** only
- **Missing or inconsistent sex disaggregation** in that particular indicator
- **Regional info being null** for these records

Would you like me to:

- Analyze this indicator using **both sexes combined**, or
- Explore a different indicator or demographic? [>-]

Envie uma mensagem para o Data Analyst